Morphophonology

Topic 5: Morpheme Structure Constraints

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So far: cyclic preservation

- A restriction/process applies/holds within a morphological subconstituent
- Process/repairs "suspended" with further affixation

A similar-looking phenomenon

- Restrictions that
  - hold at some lower level of morphological structure
  - are violated in morphologically complex words
- Result: certain configurations are banned within morphemes, but may arise in complex words

Why these are interesting

- Phonotactics and associated repairs (alternations)
    - The "duplication" problem
- Enter OT…
- Parallel vs. serial approaches

Two cases to examine

- Structure is banned within a certain class of morphemes, but found in others
  - E.g., CC clusters allowed within roots, but not affixes
- Structure is banned morpheme-internally for all classes of morphemes, but arises through morpheme concatenation

Restrictions on a class of morphemes

Lakhota (Siouan)

- CC onset clusters allowed in roots, but not prefixes

Roots		Prefixes			
blo	'potato'	wa-	1sg.subj	ka-	'by strike or wind'
mni	'water'	ma-	1sg.obj	na-	'by foot'
kte	'kill'	ja-	2nd.subj	pa-	'by hand'
tka	'scrape'	ni-	2nd.obj	wa-	'by blade'
ska	'white'	ki-	reflexive	wo-	'by piercing'
xpã	'soggy'	ki-	DAT	ja-	'by mouth'
gleʃka	'spotted'	pu-	'by pressure'	ju-	'by hands'

A widely used recipe: positional faithfulness

- General principle: faithfulness constraints may care more about 'privileged' positions
- Phonological privilege
  - Initial syllable, stressed syllable
  - Or: word-initially, before a vowel, etc.
- Morphological privilege
  - In the root
  - In nouns (Smith)

Root/affix asymmetries with root faithfulness

$/mni_{Rt}/$	Max(C)/Rt	*[CC	Max(C)
a. ☞ mni		*	
b. ni	*!		*

$/mni{+}X_{Rt}/$	Max(C)/Rt	*[CC	Max(C)
a. mni-X		*!	
b. ☞ ni-X			*

- Max(C): violated by deletion anywhere (root or affix)
- Max(C)/Rt: violated by deletion specifically in root
- Morphological affiliation of segments provided in input (same assumption as to make Align(...Stem…) work)

A pervasive phenomenon

- Lakhota: prefixes ban not just clusters, but also fricatives, aspirated stops, ejectives, laterals, nasal vowels, etc.
- Cross-linguistically: such restrictions are common
    - Inventory
    - Sequences

Side note: a puzzle about prefixes and suffixes

- Lakhota asymmetry is not actually roots vs. affixes asymmetry, but rather, prefixes vs. non-prefixes
    - -kte 'fut.', -ʃni 'neg.', -k^{h}ija 'caus.', -xtʃa 'very'
- Recall Catalan: prefixes lack a voicing contrast in stridents, but roots and suffixes do have a contrast
- In Catalan, suffixes are part of the same minimal prosodic word as the root, whereas prefixes are adjoined or separate prosodic words.
- But the same cannot be said for Lakhota…

The upshot so far

- Morphologically sensitive restrictions of the type "roots are more permissive than affixes" are unproblematic
- Roots as a privileged position for faithfulness

Restrictions on roots?

What about the converse case: affixes show greater range of structures than roots?

- Affixes have more contrasts than roots?
  - Predicted by Stratal OT: higher Faithfulness when affixes are added. (I don't know of such cases)
- Related: affixation creates more marked structures
  - Morpheme concatenation
  - Processes create marked structures in affixes

The first is far more common, so we start with that.

Morpheme-final clusters in English

English final clusters

- Morphologically simple forms: no final DD clusters
  - "DD" = two voiced obstruents (stops, fricatives)

bʌz	*bɛzd
ɹɛd	*vɛdz
lænd	*lænzd
lɛnz	*lɪndz

- Derived and inflected forms: do allow final DD

bʌzd	hɛdz
lændz	klɛnzd

Notational convention

- Fine-grained segment classes

voiceless stop	T
voiced stop	D
voiceless fric	S
voiced fric	Z
nasal	N
lateral	L
rhotic	R

- Or coarser distinctions to suit our purposes

voiceless obstruent	T
voiced obstruent	D
sonorant	R

The tip of the iceberg

- Many English final clusters occur only in affixed forms

wɪtθ	siːmd	bɛlz
fɪfθ	ɹɔŋd	kaɹz
kɔfs	læmz	wɔɹnd
liːvz	wɪŋz	sʌlkt

- Voicing assimilation and epenthesis that yield past tense and plural allomorphy may be phonology, but it's "unexpectedly little phonology"

This looks familiar! E.g., with levels…

- Stem-level: *DD] >> F

/blɪg/	*DD]	Faith
a. ☞ blɪg		
b. ?		*

/blɪgd/	*DD]	Faith
a. blɪgd	*!	
b. ☞ blɪg (blɪgəd, ...)		*

- Word-level: F >> *DD]

/blɪg+d/	Faith	*DD
a. ☞ blɪgd		*
b. blɪg	*!	

The cyclic account, in a bit more detail

- Stem level: no final DD]
- ROTB: how are they repaired?
  - Unclear, and doesn't necessarily matter
  - Assume deletion
  - Cf. *damn* ~ *damnation*, *mnemonic* ~ *amnesia*

/blɪgd/	*DD]	Dep(V)	Max(C)
a. ☞ blɪg			*
b. blɪgd	*!		
c. blɪgəd		*!	

Affixation: word level

- Need to allow final gd] created by word-level affix
- The usual gambit: faithfulness is higher
- Furthermore: Max(C) >> Dep(V)
    - Epenthesis used for certain truly bad clusters, like [COR][COR] clusters (*nidd)

/wʌg+d/	*[COR][COR]	Max(C)	Dep(V)	*DD]
a. wʌg		*!		
b. ☞ wʌgd				*
c. wʌgəd			*!	

Affixation: word level

- Newly allowable clusters

/wʌg+d/	*[COR][COR]	Max(C)	Dep(V)	*DD]
a. wʌg		*!		
b. ☞ wʌgd				*
c. wʌgəd			*!	

- Still impermissible clusters

/bad+d/	*[COR][COR]	Max(C)	Dep(V)	*DD]
a. bad		*!		
b. badd	*!			*
c. ☞ badəd			*	

An assumption of this analysis

- Crucial: -əd violates DEP
  - I.e., UR is /d/
- Some analyses rely on listed allomorphy: /d/~/t/~/əd/, but this may break the current analysis
  - -əd avoids all marked clusters
- Refraining from listing allomorphs may be safe for /-d/, /-z/, but runs into trouble with -θ/-əθ, which does seem to be phonologically conditioned allomorphy

ROTB: hypothetical suffix /-əgd/

/floʊ+gd/	*[COR][COR]	Max(C)	Dep(V)	*DD]
a. ☞ floʊgd				*
b. floʊg		*!		
c. floʊgəd			*!	

- This account doesn't really capture the generalization that DD# clusters are only created through affixation
- Captures a slightly different (incorrect) generalization: DD# clusters are permitted only in affixed words

Is there a fix for this?

- We need a level of evaluation that prohibits DD# clusters, not only in stems, but in *all* morphemes.
- Requires a slightly different cyclic architecture than what we've assumed so far
- Separate/earlier evaluation of all lexical entries

MSR's in SPR and SPE

- SPR/SPE architecture: predictable properties of morphemes provided by MSR's
- E.g., non-low back vowels are round
  - Lexicon: lacks rounding
  - MSR fill in predictable values
- How to state "no final DD"?
  - Let's assume full OT evaluation of lexical entries

We *can* import this approach, but *should* we? Is there a way of deriving it instead with the tools at hand?

A prosodic approach?

- The goal: penalize *DD in a specific prosodic domain
- Problem: actual past tense [ɹɪgd] and hypothetical but non-existent *[blɪgd] have the same prosodic structure
- Not obvious how to augment either one with additional prosodic structure

Looks impossible under this approach (???)

A transderivational approach

Two possible strategies

- BD faithfulness
  - A certain structure is banned in base forms, and faithfulness means it can't occur anywhere
  - Schematically: final DD] illegal in base, so pronounced just D] in base ⇒ D] in all forms
- Paradigm faithfulness
  - Final *DD] illegal somewhere in paradigm, and all forms must match, so banned everywhere

BD-Faithfulness: the strategy

- "Normal" application
  - Final DD] is generally banned, so repaired
- Underapplication
  - In affixed forms, available repairs are ruled out by higher-ranked constraints, so DD] is optimal

Bases: "normal" application

/blɪg/	*DD]	Dep(V)	Max(C)
a. ☞ blɪg			
b. blɪgd	*!		
c. blɪgəd		*!	

/blɪgd/	*DD]	Dep(V)	Max(C)
a. ☞ blɪg			*
b. blɪgd	*!		
c. blɪgəd		*!	

Affixed forms

For /blɪg-d/, the trick is to rule out repair candidates

/blɪg+d/ BASE: [blɪg]	???	???	*DD]	Dep(V)	Max(C)
a. blɪg	*!				*
b. ☞ blɪgd			*		
c. blɪgəd		*!		*	

Ruling out past form *[blɪg]

/blɪg+d/ BASE: [blɪg]	???	???	*DD]	Dep(V)	Max(C)
a. blɪg	*!				*
b. ☞ blɪgd			*		
c. blɪgəd		*!		*	

- Perfectly faithful to base [blɪg]
- Not just faithful stem, but identical entire string
- This obliterates the present/past distinction
  - A violation of Paradigm Contrast

Ruling out past form *[blɪg]

/blɪg+d/ BASE: [blɪg]	PC	???	*DD]	Dep(V)	Max(C)
a. blɪg	*!				*
b. ☞ blɪgd			*		
c. blɪgəd		*!		*	

- Perfectual faithful to base [blɪg]
- Not just faithful stem, but identical entire string
- This obliterates the present/past distinction
    - A violation of Paradigm Contrast

Ruling out past form *[blɪgəd]

/blɪg+d/ BASE: [blɪg]	PC	???	*DD]	Dep(V)	Max(C)
a. blɪg	*!				*
b. ☞ blɪgd			*		
c. blɪgəd		*!		*	

- [blɪg] substring looks fully faithful to base
- However, phonetically, it's quite different
  - Much shorter
  - Different realization of [g]?

Stem duration in suffixed words

Stromswold, Eisenband, Norland, and Ratzan (2002)

- Eye-tracking: comprehension of active vs. passive
  - The cow was kicking the horse
  - The cow was kicked by the horse
- Mysterious finding: adults look correctly at active scene before hearing the suffix -ing
  - Explanation: shortening in disyllabic Xɪŋ
- Effect looks small, but significant, and evidently sufficiently distinct for purposes of disambiguation
  - See also Salverda, Dahan, and McQueen (2003)

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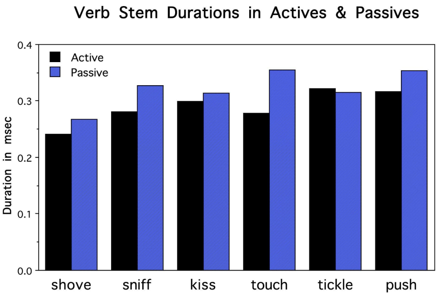

BD-Ident(dur)

/blɪg+d/ BASE: [blɪg]	PC	BD-Ident (V dur)	*DD]	Dep(V)	Max(C)
a. blɪg	*!				*
b. ☞ blɪgd			*		
c. blɪgəd		*!		*	

BD-Ident(dur)

- May be violated to satisfy higher-ranked constraints

/blɪg+ɪŋ/ BASE: [blɪg]	PC	*DN]	BD-Ident (V dur)	*DD]	Dep(V)	Max(C)
a. blɪg	*!					*
b. blɪgŋ		*!		*		
c. ☞ blɪgɪŋ			*		*	

/blɪd+d/ BASE: [blɪd]	PC	*[COR][COR]	BD-Ident (V dur)	*DD]	Dep(V)	Max(C)
a. blɪd	*!					*
b. blɪdd		*!		*		
c. ☞ blɪdəd			*		*	

Summary of the BD-Ident approach

- General restriction against clusters like -gd, -bz, etc.
- Tolerated in suffixed words only in order to keep syllable count of base and inflected form the same
- Inflection may add another syllable to satisfy even higher-ranked phonotactic constraints
- This appears to work, though it does require fine-grained faithfulness to "phonetic" duration
    - Cloud on the horizon: may not apply so easily to longer words?

A different transderivational approach

- BD-Ident: final -CC disallowed in base, and repair overapplies in derived words
- An alternative idea: final -CC disallowed somewhere in the paradigm, and repair overapplies in order to make the entire paradigm uniform
- One approach to uniform paradigms: "Optimal Paradigms" (OP: McCarthy 2005)
    - OP-Ident: violated by all mismatches within the paradigm

A paradigmatic approach

- Repair in one form extended to all forms

{/tapk/, /tapk-i/, /tapk-u/}	*CC]	OP-Dep(C)	IO-Dep(C)	IO-Max(C)
a. tapk, tapki, tapku	*!			
b. tap, tapki, tapku		*!*		*
c. ☞ tap, tapi, tapu				***

- In this particular example, if /tapk/ is a "base", BD-Dep would do the same thing

"Paradigm occultation"

- McCarthy (1998): output of /tapk/ paradigm is identical to output of /tap/ paradigm
- No evidence for learners to posit underlying contrast
- Lexicon optimization: use UR that requires as few faithfulness violations as possible
  - i.e., the UR that looks as much like the surface form as possible
- Consequence: lexicon lacks CC-final morphemes, and grammar ensures that if one were to arise, it would be repaired

Will this work for English final clusters?

- This could work in some cases
- Leading idea: final DD] clusters are tolerated, but in affixed forms they would lead to impermissible DDD]
- Repair in affixed forms overapplies in base form

Noun: {/blɪgd/, /blɪgd-z/}	*CCC]	OP-Max(C)	IO-Dep(C)	IO-Max(C)
a. blɪgd, blɪgdz	*!			
b. blɪgd, blɪgz		*!*		*
c. ☞ blɪg, blɪgz				**

This approach is brittle

- For some final CC], inflected forms would not favor deletion
  - E.g., hypothetical noun /blɪgz/
  - Plural [bligzəz] with epenthesis, not deletion of stem C
- OP depends on the paradigm a morpheme enters into, so same restrictions not guaranteed for N vs. V vs. others
  - Verbs: general phonotactics + -z, -d
  - Nouns: general phonotactics + -d
  - Adjectives: no predicted additional restrictions?
  - Also, no easy way to extend to affixes

A clever idea, that requires more complex/costlier optimization, and doesn't seem to have broad coverage for problem of MSC's

Summary of English final /-CC/ MSC

- Easy to account for by stipulating restriction, enforced on individual morphemes
  - Cost: when and why are lexical entries evaluated by phonology?
- BD-Ident approach appears to work
  - though less certainty about polysyllabic words…

Morpheme final */w/ in Koryak

Koryak (Chukotko-Kamchatkan)

Abramovitz (2021)

- Koryak has a *v~w* contrast in pre-vocalic positions

wutku	‘here’	vs.	vutqəvut	‘darkness’
waɲavatək	‘to speak’	vs.	vatqəlʔen	‘last’
ewejulʔetke	‘not scared’	vs.	ŋeveq	‘if’
təkuwiɲɲetɣi	‘I am helping you’	vs.	uvik	‘body’
mətwapaqɣele	‘we two searched for fly agaric’	vs.	mətətvamək	‘we two were’
ŋet͡ɕaqwajamək	‘in the Ngechaq river’	vs.	jaqvajat	‘which nation’

- Elsewhere, only *w* occurs

Neutralization to [w]

- v~w

waɲav-at-ə-k	‘to speak’	~	waɲaw	‘word’
jewjev-u	‘partridges’	~	jewjew	‘partridge’
t-ə-ko-pkav-ə-ŋ	‘I am unable’	~	pəkaw-ɣəjŋ-ə-n	‘inability’
in-iv-i	‘you told me’	~	t-ə-ʔ-iw-tək	‘I would tell you two’

- Invariant w

wət-at-e	‘it blossomed (of a tree)’	~	ko-wt-at-ə-ŋ	‘it is blossoming (of a tree)’
wiwət	‘whalebone’	~	wiwt-u	‘a lot of whalebone’

Grammar: contextual neutralization

- Assume that the relevant featural difference between *v* and *w* is [±son]
- Ident([±son]) penalizes changes between *v*~*w*
- *v/_¬V penalizes *v* before non-vowels

/t-ə-ʔ-iv-tk/	*v/_¬V	Ident([±son])	*w
a. ☞ t-ə-ʔ-iw-tək		*	*
b. t-ə-ʔ-iv-tək	*!		

Contextual neutralization

- Contrast before vowels

/wutku/	*v/_¬V	Ident([±son])	*w
a. ☞ wutku			*
b. vutku		*!	

/vutqvut/	*v/_¬V	Ident([±son])	*w
a. ☞ vutqəvut			
b. wutqəvut		*!	*
c. wutqəvut		*!*	**

- Neutralization to *w* elsewhere

/wiwt-u/	*v/_¬V	Ident([±son])	*w
a. ☞ wiwt-u			**
b. wivt-u	*!	*	*

/t-ə-ʔ-iv-tk/	*v/_¬V	Ident([±son])	*w
a. ☞ t-ə-ʔ-iw-tək		*	*
b. t-ə-ʔ-iv-tək	*!		

The challenge

Grammar so far predicts two types of morphemes:

- Morpheme-final /v/: surfaces as v/_V, w elsewhere
- Morpheme-final /w/: surfaces as w everywhere

/Xv-u/	*v/_¬V	Ident([±son])	*w
a. ☞ Xvu			
b. Xwu		*!	*

/Xv/	*v/_¬V	Ident([±son])	*w
a. Xv	*!		
b. ☞ Xw		*	*

/Xw-u/	*v/_¬V	Ident([±son])	*w
a. Xvu		*!	
b. ☞ Xwu			*

/Xw/	*v/_¬V	Ident([±son])	*w
a. Xv	*!	*	
b. ☞ Xw			*

⇒ In actuality, there are none like the latter

No final /w/: an accidental gap?

There seem to be reasonably many…

- Roots with prevocalic /v/ and /w/, initially, intervocalically, as C_2 in C_1C_2 clusters, etc.
- Roots with final /v/
- Affixes with internal prevocalic /v/ and /w/
- Affixes with final /v/
- Morphemes with /v/ and /w/ in clusters to show their behavior with ə epenthesis

No final /w/: a prosodic approach?

- Morphemes in question include both roots and affixes
- Expressions involving these affixes often involve CCC clusters that require epenthesis for purposes of syllabification, near or at morpheme boundaries
- If syllable boundaries coincide with PrWd boundaries, this shows that material on both sides of the morpheme boundary are in the same PrWd
- Vowel harmony also includes root and all affixes.

⇒ Unpromising to say that every morpheme is its own PrWd (or even adjoined PrWd)

No final /w/: OO faithfulness?

- All attested morphemes (roots and affixes) show alternation between w/¬_V and v/_V
- No obvious way to favor w~v over w~w with BD-Ident
    - Though cf. Catalan s~z
    - I don't know anything about the contextual phonetic realization of v and w in in Koryak
- Many of the morphemes in question are bound
    - No free-standing "base" forms to refer to

No final /w/: a cyclic approach? Stratal OT

- wV vs. vV contrastive /_V in roots and affixes
  - *v/_¬V >> Ident([±son]) >> *w at all levels
- Morpheme-final /w/
  - Must surface as [w] in output of inner cycle
  - When V-initial affix is added, will continue to be [w]
  - /w-V/ → [vV] (derived environment effect)
- Brute force *w-V ???
  - Markedness refers directly to morpheme boundary
  - Boundaries are included in surface candidates

No final /w/: morpheme-level evaluation

- Abramovitz's answer: evaluate lexical representations individually
  - Traditional MSC approach
  - Not available in all models of morphology

Constraints on morphemes

- Constraints on "alphabet" or "inventory" of UR's
  - Dispersion Theory (e.g., Flemming 2004)
  - Some MDL implementations
- Sequences: data compression? constraint evaluation?

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